

407. If $(p+3)x^2 + 4x + p = 0$ has equal roots. Find the possible values of p .

Find

408. What is the nature of the roots of $(q+1)x^2 + (2q+1)x + q = 0$?

409. If $(2k+3)x^2 - 4kx + 4 = 0$ has equal roots find the value(s) of k .

410. Find the value(s) of k for which $x^2 + (k+1)x = k+1$ has equal roots.

411. If a , b and c are three consecutive terms of an arithmetic sequence show that the equation $(b-c)x^2 + (c-a)x + a-b = 0$ has equal roots. Use the discriminant and show all your working.

Excellence Questions for Algebra



Excellence – Answer the following questions justifying solutions and giving answers in context.

412. An inventor has developed a new toy and has two marketing options, self-promotion of the product or license it to a company.

If he chooses self-promotion the return in thousands of dollars is

$$R = 9t - t^2 - 11$$

If he chooses to license it to a company then the return is

$$R = k - t$$

For what value of k will the return from the toy company always be in excess of the return from self-promotion?

Hint: Find the values for k for which the

equation $9t - t^2 - 11 = k - t$ has no solutions.

413. Find the range of values of p for which the quadratic equation

$$px^2 + (p+3)x + p = 0$$

has real roots.

414. If $p(x)$ is a polynomial and is divided by $(x-k)$ and a remainder is obtained, then that remainder is $p(k)$.

If the quadratic $p(x) = x^2 - 3x + 5$ gives the same remainder when divided by $x+k$ as it does when divided by $x-3k$, find the value of k , $k \neq 0$.

415. Find the range of values of q for which the polynomial is always negative.

$$f(x) = 2qx^2 + 2(q+4)x + q + 1$$

416. Make x the subject of the expression

$$y^{(4x+1)} = 100y^x$$

Simplify your answer as much as possible. For what values of y does your answer exist for? Justify any restrictions.

417. The parabola $y = 2x^2 - 8x + 4$ intersects the straight line $y = 4x + k$ once only (i.e. it is a tangent). Using algebra find the value of k for this to be true and the coordinates of the point of intersection.

418. Josh's family own some land on Waiheke Island and he has watched it increase in value.



Josh believes that the family's land on Waiheke will make him rich. He estimates that when it is worth 25 times his family's income they should sell it and live off the money for the next 25 years.

His mother is concerned that Josh has not taken inflation into account. The family income now is \$68 000 and it is increasing at 3 percent per year as given by the formula

$$\text{Income}(000) = 68 (1.030)^N$$

(where N is the number of years into the future). The land is valued at \$500 000 now and is increasing as shown by

$$\text{Value}(000) = 500 (1.095)^N$$

(where N is the number of years into the future).

They purchased it for \$50 000 in 1980 and it has increased in value by 9.5% per year since.

Josh wants to know when it is likely to be worth five million dollars.

- a) Solve the equation

$$\text{Value}(\$ 000) = 50 (1.095)^T$$

where T is the number of years since 1980 and the Value is in \$1000s. Form and solve an equation to find when it will be worth five million dollars.

- b) Calculate when the value will be 25 times the present family income. Show all equations and working.

419. According to the specifications that came with Laura's phone the signal strength is given by
- $$S = k(d + 0.5)^{-1.2}$$

where S is a figure from 0 to 100 (100%), k is a constant and d (in km) is the distance from the mobile phone tower.

When the phone is near the tower ($d = 0$) the signal strength is 100% ($S = 100$).

- a) How far from the tower will Laura be when the signal strength drops to 5% ($S = 5$)?



Laura lives in Hong Kong but likes to send text messages to her friend back in New Zealand. She has found that after receiving a message, the length of time it takes her to reply depends upon the time to enter the message and the transmission time.

The time to enter the message with c characters is given by $t = 0.45c$ where t is in seconds and c is the number of characters.

The transmission time is always a minimum of 5 seconds and the rest of the transmission time is best modelled by the equation $0.015c^2$ where again t is in seconds after the 5 second delay and c is the number of characters.

It takes Laura 27.1 seconds to enter and send a text back to her friend in New Zealand.

- c) How many characters are there in her text? You will need to form an equation and solve it to justify your answer.

The rechargeable battery in the phone is designed to last 5 days (120 hours). After about thirteen weeks it will start to deteriorate and the length of time in hours between charges is modelled by the equation

$$L = 120 \cdot 1.0151^{(t/3-4)} \quad (t \geq 13)$$

where L is the life in hours and t is the time in weeks since purchase. Laura will change her phone when the battery lasts 48 hours or less.

- b) How long will Laura have the phone? You must form an equation and solve it.

Practice External Assessment Task

Algebraic Methods 2.6

Make sure you show ALL relevant working for each question. You are advised to spend 60 minutes answering this assessment.

QUESTION ONE

- (a) Expand and simplify $(2x + 5)(x - 3)(x - 2)$

- (b) Write as a single log statement $3 \log A + \log B - 0.5 \log C$

- (c) i) Solve the equation $3x^2 - 4x + 1 = 0$

- ii) For what value of k does the equation $3x^2 + kx + 4 = 0$ have equal roots?

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302. a) Long side x and short side $x - 12$
 b) Long side $x + 30$ and short side $x + 18$
 c) $2(2x + 2(x - 12)) = 2(x + 18) + 2(x + 30)$
 or equivalent
 d) $x = 36$
 Length 36 m
 Width 24 m
 303. a) There are 20 who support the idea (19 + 1) and 24 + x who were at the meeting.

$$\frac{1}{x}$$

$$\frac{20}{24 + x} - \frac{1}{x} = \frac{1}{2}$$

$$38x - 48 = x(24 + x)$$

$$x = 6, 8$$

- d) Two answers and both possible.

$$\frac{20}{24 + x} - \frac{1}{x} = \frac{1}{2}$$

$$\frac{30}{30 - 6} = \frac{1}{2}$$

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304. a) $1500x + 800(2x + 3) + 4800x$
 b) $1.5x + 0.8(2x + 3) + 4.8x = 20$
 $x = 2.228$

Order: 2 m³ cement,
 (4 + 3) = 7 m³ sand and
 6 m³ crushed stone.

$$305. \text{ a) Energy} = 11t + 2t \times 5$$

$$\text{b) Energy} = 21t + (60 - 3t) \times 8$$

$$\text{c) } 21t + 8(60 - 3t) = 440$$

Jogging 13 minutes,
 walking 27 minutes and
 Cycling 20 minutes.

Answers to 0 dp.

Time cafe = t

$$5(13.5t + 2t \times 10 + 20) = 600$$

$$t = 2.985$$

Cafe = 3 hours (0 dp)

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$$307. x < 9$$

$$308. x < 4$$

$$309. r \leq 7\frac{1}{2} (7.5)$$

$$310. k < 10$$

$$311. x < 7$$

$$312. x < 4\frac{1}{4} (4.25)$$

$$313. k \geq 5\frac{1}{4} (5.25)$$

$$314. m \geq -4\frac{1}{2} (-4.5)$$

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350. $x = -11, 2$
 351. $x = 3$
 352. $x = \frac{1}{2}, \frac{5}{2}$
 353. $x = -1, \frac{1}{6}$
 354. $x = -1\frac{1}{2}, \frac{1}{4}$
 355. $x = 3\frac{1}{2}, -1\frac{1}{2}$
 356. $x = -5, 5$
 357. $k = -1\frac{1}{2}, \frac{1}{2}$
 358. $x = \pm\sqrt{3}, (\pm 1.732)$ only

$$359. x = \frac{1}{25} (0.04) \text{ only}$$

$$360. x = \frac{1}{2} \text{ only as } x = 1$$

involves $\div$ by 0.

$$361. b = \pm 1, \pm\sqrt{\frac{5}{4}} (\pm 1.118)$$

$$362. z = 3 \text{ only}$$

$$363. p = \sqrt[3]{3}, \sqrt[3]{\frac{6}{5}} \text{ or } p = 1.4422, -1.0627 (4 \text{ dp})$$

$$364. \text{ a) } (x + 18)(x + 12) = 432$$

$$\text{b) } x^2 + 30x - 216 = 0$$

$$\text{c) } (x - 6)(x + 36) = 0$$

$$\text{d) } x = 6 \text{ m only}$$

$$x = -36 \text{ is rejected}$$

$$365. \text{ a) } (20 - x)(15 - x) = 249.75$$

$$x^2 - 35x + 50.25 = 0$$

$$4x^2 - 140x + 201 = 0$$

$$\text{b) } (2x - 3)(2x - 67)$$

$$x = 1.5, 33.5$$

$$\text{Ignore } 33.5$$

$$\text{Length} = 18.5 \text{ m}$$

$$\text{Width} = 13.5 \text{ m}$$

$$366. \text{ a) } \begin{array}{|c|c|} \hline x & x+8 \\ \hline \end{array}$$

$$x^2 + 14x - 207 = 0$$

$$(x + 23)(x - 9) = 0$$

$$x = 9, -23$$

Ignore -23

$$\text{Width} = 9 \text{ m}$$

$$\text{Length} = 17 \text{ m}$$

$$\text{b) } 2x(x + 11) = x(x + 8)$$

$$x^2 + 14x - 207 = 0$$

$$(x + 23)(x - 9) = 0$$

$$x = 9, -23$$

Ignore -23

$$\text{Width} = 9 \text{ m}$$

$$\text{Length} = 17 \text{ m}$$

$$\text{c) } x^2 + 14x - 207 = 0$$

$$(x + 23)(x - 9) = 0$$

$$x = 9, -23$$

Ignore -23

$$\text{Width} = 9 \text{ m}$$

$$\text{Length} = 17 \text{ m}$$

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391. $\Delta = 0$ one rational root
 392. $\Delta = 16$ two rational roots
 393. $\Delta = -8$ no real roots
 394. $\Delta = 0$ one rational root
 395. $\Delta = 49$ two rational roots
 396. $\Delta = 37$ two irrational roots
 397. $\Delta = 0$ one rational root
 398. $\Delta = 0.96$ two irrational roots
 399. $\Delta = 169$ two rational roots
 400. $\Delta = 8$ two irrational roots
 401. $b^2 - 64 = 0$
 $b = 8$ or -8

Lengths 9 cm, 12 cm, 15 cm

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369. $x = 4.562, 0.438$ (3 dp)
 370. $x = -0.228, -8.772$ (3 dp)
 371. $x = -1, -1.5$
 372. $x = 1.212, -3.712$ (3 dp)
 373. $x = -0.697, -4.303$ (3 dp)
 374. $x = 0.425, -1.175$ (3 dp)

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375. $x = 0.309, -0.809$ (3 dp)
 376. $x = -0.775, -3.225$ (3 dp)
 377. $x = -1.5$
 378. $x = -0.469, -8.531$ (3 dp)
 379. $m = 1.644, -0.553$ (3 dp)
 380. $y = 2.610, -1.277$ (3 dp)
 381. $x = 1.844, -1.844$ (3 dp)
 382. $x = 4, -1.5$
 383. $p = -0.057, 2.182$ (3 dp)
 384. $x = -2.5, 2.333$ (3 dp)
 385. $a = 7, -0.5$
 386. $q = 2.443, -4.776$ (3 dp)
 387. $x = -2, 2$
 388. $x = -0.215, 1.549$ (3 dp)

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407. $16 - 4p(p + 3) = 0$
 $4p^2 + 12p - 16 = 0$
 $4(p + 4)(p - 1) = 0$
 $p = -4, 1$
 408. $\Delta = (2q + 1)^2 - 4q(q + 1)$
 $\Delta = 1$ two rational roots
 409. $16k^2 - 4(2k + 3)(4) = 0$
 $16k^2 - 32k - 48 = 0$
 $16(k - 3)(k + 1) = 0$
 $k = 3, -1$
 410. $(k + 1)^2 - 4(1)(k - 1) = 0$
 $k^2 + 6k + 5 = 0$
 $(k + 5)(k + 1) = 0$
 $k = -5, -1$
 e) Solve for x to find increase = 40¢ so Price = \$1.10

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390. a) $x + 12$ and $x + 14$
 b) $(x + 14)^2 = (x + 12)^2 + x^2$
 c) $x^2 - 4x - 52 = 0$
 d) $(x - 2)^2 - 56 = 0$
 $x = -5.5, 9.5$ (2 sf)
 Length of sides to 2 sf.
 are 9.5, 21.5 and 23.5 m.

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412. $t^2 - 10t + 11 + k$
 Discriminant < 0
 $100 - 44 - 4k < 0$
 $k > 14$
 413. $(p + 3)^2 - 4(p)(p) \geq 0$
 $-3p^2 + 6p + 9 \geq 0$
 $-3(p - 3)(p + 1) \geq 0$
 $-1 \leq p \leq 3$

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414. $p(k) = k^2 + 3k + 5$
 $p(3k) = 9k^2 - 9k + 5$
 Equating
 $k^2 + 3k + 5 = 9k^2 - 9k + 5$
 $8k^2 - 12k = 0$
 $4k(2k - 3) = 0$
 Ignoring $k = 0$ so $k = 1.5$
 415. Discriminant must be negative for no roots and below axes so 2q must be negative.
 $0 > 4(q + 4)^2 - 4(2q)(q + 1)$
 $0 > 4q^2 + 32q + 64 - 8q^2 - 8q$
 $0 > -4q^2 + 24q + 64$
 $0 < 4q^2 - 24q - 64$
 $0 < q^2 - 6q - 16$
 $0 < (q - 8)(q + 2)$
 This is true if $q > 8$ OR $q < -2$.
 As 2q must be negative then answer $q < -2$.

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416. Take logs base 10 of both sides.
 $\log(y^{(4x+1)}) = \log(100y^x)$
 $(4x + 1)\log y = \log 100 + x\log y$
 $(4x + 1)\log y = \log 100 + x\log y$
 $4x\log y - x\log y = 2 - \log y$
 $x(4\log y - \log y) = 2 - \log y$
 $x = \frac{2 - \log y}{3\log y}$
 and $y > 0$, but $y \neq 1$.
 When $y > 0$ log y exists, but if $y = 1$, log 1 = 0 and we cannot divide by zero.

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417. Substitute for y in the parabola
 $y = 2x^2 - 8x + 4$
 $4x + k = 2x^2 - 8x + 4$
 $0 = 2x^2 - 12x + 4 - k$
 $a = 2, b = -12, c = 4 - k$
 $144 - 32 + 4 - k = 0, k = 14$
 $0 = 2x^2 - 12x + 18$
 Intersection is (3, -2).

